Trends in mortality from cancers of the breast, colon, prostate, esophagus, and stomach in East Asia: role of nutrition transition.

Although substantial nutrition transition, characterized by an increased intake of energy, animal fat, and red meats, has occurred during the last several decades in East Asia, few studies have systematically evaluated temporal trends in cancer incidence or mortality among populations in this area. Therefore, we sought to investigate this question with tremendous public health implications. Data on mortality rates of cancers of the breast, colon, prostate, esophagus, and stomach for China (1988-2000), Hong Kong (1960-2006), Japan (1950-2006), Korea (1985-2006), and Singapore (1963-2006) were obtained from WHO. Joinpoint regression was used to investigate trends in mortality of these cancers. A remarkable increase in mortality rates of breast, colon, and prostate cancers and a precipitous decrease in those of esophageal and stomach cancers have been observed in selected countries (except breast cancer in Hong Kong) during the study periods. For example, the annual percentage increase in breast cancer mortality was 5.5% (95% confidence interval: 3.8, 7.3%) for the period 1985-1993 in Korea, and mortality rates for prostate cancer significantly increased by 3.2% (95% confidence interval: 3.0, 3.3%) per year from 1958 to 1993 in Japan. These changes in cancer mortality lagged ~ 10 years behind the inception of the nutrition transition toward a westernized diet in selected countries or regions. There have been striking changes in mortality rates of breast, colon, prostate, esophageal, and stomach cancers in East Asia during the last several decades, which may be at least in part attributable to the concurrent nutrition transition.